# Cross-Generational Transfer of Adversarial Attacks Reveals Non-Monotonic Safety Alignment in LLMs

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## Abstract

Safety alignment in LLMs does not improve monotonically across model generations. Studying four generations of Google's Gemma family (7B–31B) with quality-diversity evolution (MAP-Elites) as an automated red-teaming probe, we find that Gemma 3 (12B) exhibits $68.7\% \pm 5.7\%$ attack success rate (ASR; mean $\pm$ std, 3 seeds), significantly higher than its predecessor Gemma 2 ($45.5\% \pm 7.2\%$; $p = 0.030$, paired bootstrap) and its successor Gemma 4 ($33.9\% \pm 1.8\%$). Replaying evolved attack archives across generations reveals that attacks from *other* generations transfer to Gemma 3 at 44–46% but only 14–18% to Gemma 4, indicating that Gemma 4's safety gains generalize beyond the attack distributions evolved against earlier generations. Under our 8B judge, copyright and cybercrime vulnerabilities register at near-100% across all generations, though a second-judge audit (§6) suggests the copyright result is sensitive to judge choice. Misinformation ASR jumps from 29% to 99% between Gemma 2 and Gemma 3 and remains elevated at 77% in Gemma 4, indicating the regression was not fully addressed. These patterns are invisible to static benchmarks and emerge only through adaptive, longitudinal probing. All experiments use 3 random seeds with a unified self-hosted judge; code and artifacts are available anonymously.

## 1 Introduction

Large language model (LLM) safety alignment has improved rapidly, but measuring that improvement rigorously remains an open challenge. Static benchmarks such as HarmBench Mazeika et al. [2024] and AdvBench Zou et al. [2023] provide point-in-time snapshots but do not capture how an adaptive adversary interacts with evolving defenses.

A fundamental question is whether safety improvements *generalize*: when a model developer patches vulnerabilities between releases, do old attacks stop working (indicating targeted fixes), or does the model become broadly more robust? Answering this requires a methodology for *differential safety probing*: systematically comparing safety surfaces across model versions with a consistent adversarial process.

In this paper, we address two specific questions:

1. **Cross-generational transfer:** Do adversarial attacks evolved against an earlier model generation transfer to later generations of the same family?
2. **Non-monotonic robustness:** Does safety alignment improve consistently across model generations, or can it regress?

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

We study these questions using four generations of Google's Gemma model family (7B, 9B, 12B, 31B parameters), spanning releases from 2024 to 2026. Our probe is a quality-diversity (QD) evolutionary algorithm (MAP-Elites) that discovers diverse jailbreak strategies across seven harm categories from the HarmBench taxonomy. By evolving attack archives against each model with 3 independent seeds and replaying them across generations, we construct a transfer matrix with uncertainty estimates that reveals which vulnerabilities persist and which are patched.

**Contributions.**

- A QD-evolution methodology for *differential safety probing* that enables systematic, reproducible comparison of safety surfaces across model versions (§3).
- A cross-generational transfer analysis across four Gemma generations, revealing that attack transfer is generation-dependent and that Gemma 3 represents a statistically significant safety regression ($p = 0.030$; §5.2).
- Empirical evidence that adversarial robustness does not improve monotonically across model generations within a single family, with a non-monotonic pattern coinciding with an architectural transition (§5.3).
- Category-level analysis showing that some harm categories (notably copyright, under our 8B judge) appear to resist safety alignment across all generations while others show broadly declining trajectories (§6).

## 2 Related Work

**Automated red-teaming.** Recent work has developed automated methods for discovering LLM vulnerabilities, including gradient-based attacks (GCG; Zou et al. [2023]), LLM-based attackers (PAIR Chao et al. [2023]; TAP Mehrotra et al. [2024]), and template-based approaches (AutoDAN Liu et al. [2024]). These methods target individual models; none study how discovered vulnerabilities transfer across model generations.

**Safety benchmarks.** HarmBench Mazeika et al. [2024] provides a standardized taxonomy of harmful behaviors across seven categories. StrongREJECT Souly et al. [2024] proposes improved evaluation metrics. We use HarmBench as our behavior taxonomy and seed pool, enabling category-level transfer analysis.

**Quality-diversity optimization.** MAP-Elites Mouret and Clune [2015] maintains a grid of diverse, high-performing solutions across behavioral dimensions. Prior work has applied QD methods to game playing Fontaine et al. [2020] and robot design Cully et al. [2015]. Anonymous Anonymous [2026] established that MAP-Elites discovers more diverse LLM vulnerabilities than standard evolutionary search, in a single-model setting. We adopt their probe and extend it in three directions: a cross-generational transfer replay protocol, a four-generation longitudinal study with three seeds per model, and the empirical finding that safety alignment regresses non-monotonically across Gemma generations.

**Scaling laws.** Scaling laws for LLM capabilities are well-established Kaplan et al. [2020], Hoffmann et al. [2022]. Whether safety properties follow similar predictable scaling is largely unexplored. We provide evidence that adversarial robustness does *not* scale monotonically within a single model family, a finding that contrasts with the regularity observed in capability scaling.

## 3 Methodology

### 3.1 Quality-Diversity Adversarial Probing

We use MAP-Elites to evolve a diverse archive of jailbreak attacks against a target LLM. Each attack is characterized by a genome encoding:

- **Attack strategy** (6 types): roleplay, encoding, authority impersonation, hypothetical framing, multi-turn, and direct jailbreak.

Table 1: Models evaluated. All served via vLLM 0.19.1 (bf16, temperature 0) on a single NVIDIA H200 (141 GB). Judge: self-hosted Llama-3.1-8B-Instruct on NVIDIA RTX 4090 (24 GB).

| Model | Family | Params | Release | Architecture |
|---|---|---|---|---|
| Gemma 1 (7B-it)[a] | Gemma | 7B | 2024 | Text-only |
| Gemma 2 (9B-it)[b] | Gemma | 9B | 2024 | Text-only |
| Gemma 3 (12B-it)[c] | Gemma | 12B | 2025 | Multimodal |
| Gemma 4 (31B-it)[d] | Gemma | 31B | 2026 | Multimodal |

[a]`google/gemma-7b-it` [b]`google/gemma-2-9b-it` [c]`google/gemma-3-12b-it` [d]`google/gemma-4-31B-it`

- **Encoding** (6 types): none, Base64, ROT13, leetspeak, pig latin, Unicode substitution.
- **Seed behavior**: drawn from HarmBench's 400 behaviors across 7 semantic categories.

The MAP-Elites archive is a 3D grid indexed by (strategy $\times$ encoding $\times$ harm category), with each cell storing the highest-fitness attack for that behavioral niche. Fitness is evaluated by a self-hosted LLM judge (Llama-3.1-8B-Instruct) that classifies each target response along three binary axes: refusal, harmful content, and relevance. The resulting score ranges from 0 (refused) to 1 (harmful and relevant). Using a single, self-hosted judge for *all* phases (evolution and transfer replay) ensures scoring consistency and full reproducibility.

Each model is probed with 3 independent random seeds (42, 1337, 2718) to estimate cross-seed variance. All evolution uses deterministic RNG (ChaCha8, seeded) with ordered evaluation to ensure reproducibility given the same seed.

### 3.2 Cross-Generational Transfer Protocol

After evolving an archive $\mathcal{A}_{i,s}$ against model $M_i$ with seed $s$, we *replay* each archive against all other models $M_j$ ($j \neq i$):

1. For each entry in $\mathcal{A}_{i,s}$, send the attack prompt to $M_j$ and obtain the response.
2. Score the response using the same LLM judge.
3. Record per-entry and per-category transfer success.

This produces a transfer tensor $T[i, j, s]$ where each entry is the fraction of attacks from $\mathcal{A}_{i,s}$ that succeed against $M_j$. We report the mean and standard deviation across seeds. The diagonal $\bar{T}[i, i]$ is the mean self-evaluation success rate. Off-diagonal entries reveal:

- **Forward transfer** ($i < j$): old attacks on newer models. High values suggest safety improvements are superficial.
- **Backward transfer** ($i > j$): newer attacks on older models. High values suggest attacks exploit longstanding weaknesses.

### 3.3 Robustness Across Generations

We plot robustness $R = 1 - \text{ASR}$ as a function of model generation (with parameter count on the axis) to visualize the trajectory of safety alignment within the Gemma family. We treat the four generations as a case study and test specific pairwise contrasts with paired bootstrap tests ($p < 0.05$, uncorrected; $N = 3$ seeds limits power for multiple-testing correction).

## 4 Experimental Setup

### 4.1 Models

We note that Gemma 3 and Gemma 4 introduced multimodal (vision-language) capabilities, representing an architectural shift from the text-only Gemma 1 and Gemma 2. This distinction is relevant to interpreting the safety regression observed in Gemma 3 (§6).

Table 2: Attack success rate (%) per HarmBench category across Gemma generations (mean ± std, 3 seeds). **Bold** indicates lowest (safest) ASR per category. Cells with ±0.0 reflect floor or ceiling effects where all seeds converged to the same value.

| Category | Gemma 1 (7B) | Gemma 2 (9B) | Gemma 3 (12B) | Gemma 4 (31B) |
|---|---|---|---|---|
| Chem/Bio | $40.7 \pm 6.4$ | $31.5 \pm 1.6$ | $54.6 \pm 30.0$ | $\mathbf{0.0 \pm 0.0}$ |
| Copyright | $100.0 \pm 0.0$ | $100.0 \pm 0.0$ | $100.0 \pm 0.0$ | $100.0 \pm 0.0$ |
| Cybercrime[†] | $97.2 \pm 4.8$ | $72.2 \pm 41.1$ | $100.0 \pm 0.0$ | $48.1 \pm 44.5$ |
| Harassment | $33.3 \pm 0.0$ | $22.2 \pm 4.8$ | $28.7 \pm 5.8$ | $\mathbf{1.9 \pm 1.6}$ |
| Harmful | $72.2 \pm 23.6$ | $\mathbf{26.4 \pm 2.0}$ | $34.7 \pm 9.8$ | $\mathbf{0.0 \pm 0.0}$[‡] |
| Illegal | $44.4 \pm 19.2$ | $33.3 \pm 28.9$ | $50.0 \pm 17.3$ | $\mathbf{0.0 \pm 0.0}$ |
| Misinfo | $33.3 \pm 0.0$ | $\mathbf{28.7 \pm 1.6}$ | $99.1 \pm 1.6$ | $76.9 \pm 25.8$ |
| **Overall** | $59.4 \pm 3.0$ | $45.5 \pm 7.2$ | $68.7 \pm 5.7$ | $\mathbf{33.9 \pm 1.8}$ |

[†]Gemma 4 cybercrime shows high cross-seed variance (2.8%, 50.0%, 91.7% across seeds), reflecting genuine sensitivity to the evolutionary trajectory rather than a pipeline error.

[‡]Based on 2 seeds; one seed's archive contained zero harmful-category entries (the evolution did not populate those niches).

### 4.2 Evolution Configuration

MAP-Elites with $6 \times 6 \times 8$ grid (strategy × encoding × category), yielding 288 behavioral niches. Batch size 50, 200 generations per model per seed, concurrency 5 parallel evaluations. Mutation rate 0.3, crossover rate 0.7. Seed pool: 400 HarmBench behaviors across 7 categories. Three seeds per model (42, 1337, 2718) yield 12 evolution runs total. Transfer replay: 36 source–target pairs (4 models × 3 other targets × 3 seeds).

### 4.3 Compute

All Gemma models served sequentially on a single NVIDIA H200 (141 GB HBM3e) via vLLM 0.19.1. LLM judge (Llama-3.1-8B-Instruct) served on a dedicated NVIDIA RTX 4090 (24 GB). Total GPU time: approximately 60 hours across both machines (12 evolution runs at ∼4 hours each, 36 transfer replays at ∼8 minutes each, plus 4 raw-seed baselines and a 70B judge audit).

## 5 Results

### 5.1 Baseline Attack Success Rates

Table 2 presents attack success rates (ASR) across all HarmBench categories for each Gemma generation, reported as mean ± standard deviation across 3 seeds. The most striking observation is the *non-monotonic* pattern: overall ASR decreases from $59.4\% \pm 3.0\%$ (Gemma 1) to $45.5\% \pm 7.2\%$ (Gemma 2), then *increases* to $68.7\% \pm 5.7\%$ (Gemma 3), before dropping to its lowest value of $33.9\% \pm 1.8\%$ (Gemma 4).

The improvement from Gemma 1 to Gemma 2 ($p = 0.032$) and the regression from Gemma 2 to Gemma 3 ($p = 0.030$) are both statistically significant under paired bootstrap tests. The recovery from Gemma 3 to Gemma 4, while large in absolute terms ($-34.8$ percentage points), does not reach significance at $p < 0.05$ with 3 seeds, a limitation of our sample size that we discuss in §6.

A raw-seed baseline (400 HarmBench prompts, no evolution) confirms that evolution adds 7–14 percentage points across models and that the non-monotonic pattern holds even without evolution.

Under our 8B judge, copyright is consistently the highest-ASR category, though a second-judge audit (§6) suggests the 8B judge may over-classify copyright responses as harmful. Gemma 4 achieves 0% ASR in chem/bio, harmful, and illegal, but misinformation ($76.9\% \pm 25.8\%$) and cybercrime ($48.1\% \pm 44.5\%$) remain high. Misinformation spikes from $28.7\%$ in Gemma 2 to $99.1\%$ in Gemma 3 and stays elevated at $76.9\%$ in Gemma 4. Harassment shows a broadly declining trajectory ($33.3\% \rightarrow 22.2\% \rightarrow 28.7\% \rightarrow 1.9\%$), supporting the interpretation that category-specific safety training was maintained even when general robustness regressed.

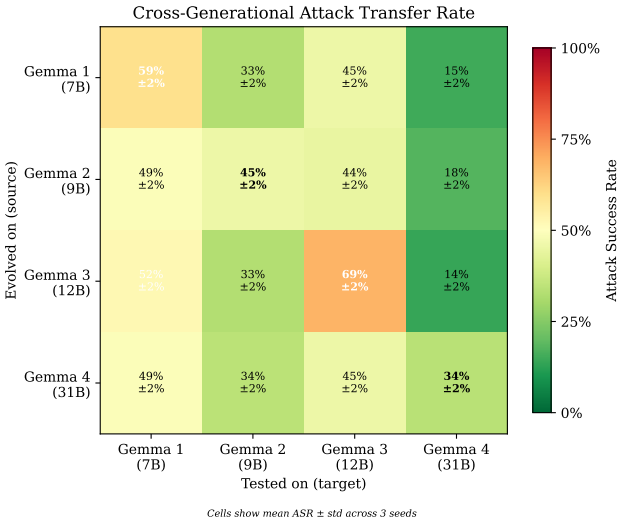



Figure 1: Cross-generational attack transfer matrix (mean $\pm$ std across 3 seeds). Cell $(i, j)$ shows the fraction of attacks evolved against Gemma $i$ that succeed on Gemma $j$. Diagonal entries (bold) are self-evaluation rates.

### 5.2 Cross-Generational Transfer

Figure 1 presents the full $4 \times 4$ transfer matrix with uncertainty estimates. Several patterns emerge:

**Gemma 3 is the easiest transfer target.** Archives from other generations transfer to Gemma 3 at 44–46% ($45.5\%$/$44.5\%$/$45.5\%$ from Gemma 1/2/4). A niche decomposition finds Jaccard similarity of 0.66–0.75 among the successful niches across source archives, indicating substantial overlap in which (strategy $\times$ encoding $\times$ category) cells succeed regardless of source.

**Gemma 4 resists transfer.** Transfer rates to Gemma 4 are 14–18% from all sources. Despite Gemma 3's high self-ASR ($68.7\%$), its attacks do not generalize to Gemma 4, suggesting Gemma 4's improvements generalize beyond the specific attack distributions evolved against earlier generations.

**Backward transfer is substantial.** Attacks from all generations transfer to Gemma 1 at $\sim$49–52%, indicating that newer probes discover longstanding vulnerabilities.

### 5.3 Robustness Across Generations

Figure 2 plots ASR and robustness ($1 - \text{ASR}$) against parameter count with error bars. We do not fit a scaling law to four data points. Instead, we observe that three of the four models (Gemma 1, 2, 4) are consistent with a gradual improvement trajectory, while Gemma 3 represents a clear outlier. This non-monotonic pattern coincides with the architectural transition from text-only (Gemma 1, 2) to multimodal (Gemma 3, 4), though Gemma 4's recovery shows that the regression is not an inherent consequence of multimodal design. Gemma 4's low overall ASR is driven primarily by near-zero ASR in three categories (chem/bio, harmful, illegal); misinformation ($76.9\%$) and cybercrime ($48.1\%$) remain high.

### 5.4 Vulnerability Fingerprints

Figure 3 presents vulnerability fingerprints as radar charts. Each model exhibits a distinct profile:

- **Gemma 1**: moderate vulnerability with peaks in cybercrime ($97.2\%$) and copyright ($100\%$).
- **Gemma 2**: the most compact profile overall, though copyright remains at 100% and cybercrime at $72.2\%$.

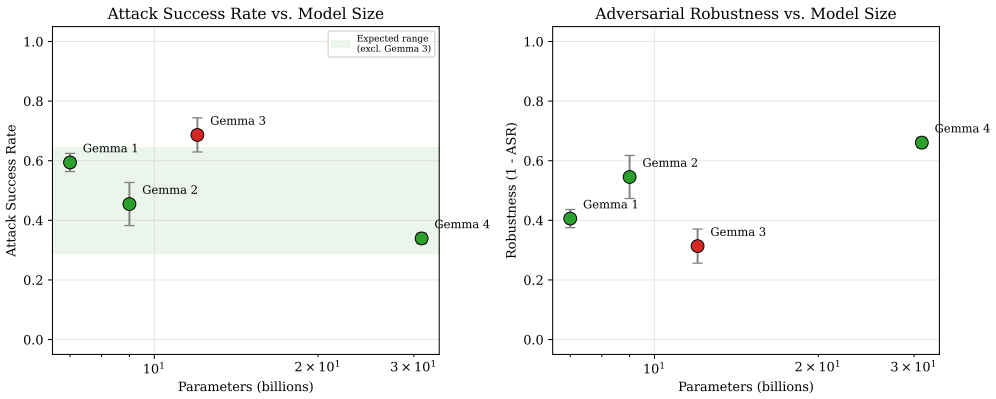



Figure 2: Attack success rate (left) and adversarial robustness (right) vs. model size with error bars ($\pm$ 1 std, 3 seeds). The green shaded region shows the range spanned by Gemma 1, 2, and 4; Gemma 3 (red) falls well outside this range, demonstrating the non-monotonic pattern. No scaling law is fit; four data points are insufficient.

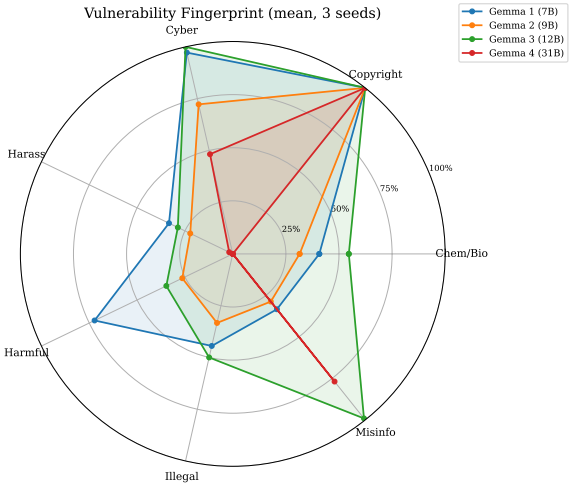



Figure 3: Vulnerability fingerprints across Gemma generations (mean ASR per category, 3 seeds). Gemma 2 and Gemma 4 show compact profiles; Gemma 3's profile is dramatically expanded in cybercrime and misinformation.

- **Gemma 3**: dramatically expanded in misinformation (99.1%) and cybercrime (100%). Only harassment shows continued improvement.
- **Gemma 4**: achieves 0% ASR in chem/bio, harmful, and illegal, but copyright (100%) and misinformation (76.9%) remain elevated.

## 6 Discussion

**Safety regression in Gemma 3.** The most surprising finding is Gemma 3's safety regression, which is statistically significant versus Gemma 2 ($p = 0.030$, paired bootstrap across 3 seeds). We hypothesize this is related to the introduction of multimodal capabilities: Gemma 3 was the first Gemma generation with vision-language support (architecture: `Gemma3ForConditionalGeneration`), requiring substantial architectural changes that may have disrupted safety-trained text representations. The fact that harassment, the category most explicitly targeted by safety training across all LLMs, shows a broadly declining trajectory even through Gemma 3 ($33.3\% \rightarrow 22.2\% \rightarrow 28.7\% \rightarrow 1.9\%$) supports this interpretation: category-specific safety training was maintained, but general robustness was compromised. Alternative hypotheses that the data does not rule out include (a) idiosyncratic

training-data-mix issues specific to Gemma 3 unrelated to modality, (b) changes in the RLHF/DPO recipe between generations, and (c) increased model capability raising the ceiling on evolutionary search. Distinguishing these would require access to training details, which we do not have.

**Gemma 4's recovery.** Gemma 4 achieved the lowest overall ASR in the family ($33.9\% \pm 1.8\%$) despite being multimodal, suggesting that the safety regression in Gemma 3 was addressable. The extremely low transfer rate from Gemma 3 to Gemma 4 ($14.0\% \pm 2.7\%$) indicates that Gemma 4's improvements generalize beyond the specific attack distributions evolved against Gemma 3. However, Gemma 4's safety is category-heterogeneous: three categories reach 0% ASR while misinformation ($76.9\%$) and cybercrime ($48.1\%$) remain high.

**Judge sensitivity and copyright.** Copyright-category ASR is consistently highest across all generations under our 8B judge. However, a second-judge audit using Llama-3.1-70B-Instruct ($N = 52$ samples with stored responses) found that the 70B judge classified only 50% of copyright responses as harmful (vs. 100% for the 8B judge), suggesting that the 8B judge's harm threshold for copyright is more permissive. Overall, the two judges agree on refusal (96.2%) but show weak agreement on harm classification ($\kappa = 0.15$). The second-judge audit covered 52 samples, which limits per-category resolution; a larger audit would be needed to characterize judge disagreement per category. Per-category ASRs should be interpreted with this caveat.

**What we claim and what we do not.** With four data points we cannot distinguish a U-shaped curve from noise. What we *can* say is that the Gemma 2 $\rightarrow$ Gemma 3 regression is statistically significant and large ($+23.2$ percentage points), and that the transfer matrix provides converging evidence (Gemma 3 is the easiest transfer target from all directions).

**Implications for safety evaluation.** Static benchmarks would miss the non-monotonic trajectory entirely. Architectural changes (such as adding multimodal capabilities) should trigger comprehensive safety re-evaluation. The distinction between Gemma 3 (high self-ASR, high inbound transfer) and Gemma 4 (low self-ASR, low inbound transfer) is only visible through cross-generational transfer analysis.

**Limitations.**

- Three seeds per model provide reproducibility estimates but limited statistical power. The Gemma 3 $\rightarrow$ Gemma 4 improvement, while large ($-34.8$ pp), does not reach $p < 0.05$. Additional seeds would tighten confidence intervals.
- Single model family (Gemma) for generational analysis; findings may not generalize to other families.
- The 8B judge shows weak agreement ($\kappa = 0.15$) with a 70B judge on harm classification, particularly for copyright. Per-category ASRs are judge-dependent.
- HarmBench coverage: 400 behaviors across 7 categories is comprehensive but not exhaustive.
- Evolutionary probe explores a subset of the attack space; absence of a successful attack does not prove safety.
- We study only the text modality; Gemma 3 and 4's vision capabilities may present additional attack surfaces not captured here.

# 7 Conclusion

We have presented a methodology for longitudinal safety evaluation of LLMs using quality-diversity evolution as a differential probe. Applied to four generations of Google's Gemma models with 3 independent seeds per model, we find that:

1. Safety alignment does not improve monotonically across model generations. Gemma 3 represents a statistically significant regression ($68.7\% \pm 5.7\%$ ASR vs. Gemma 2's $45.5\% \pm 7.2\%$; $p = 0.030$).

2. Cross-generational attack transfer is generation-dependent: attacks from other generations transfer to Gemma 3 at $\sim$45% but only 14–18% to Gemma 4, suggesting Gemma 4's improvements generalize across attack distributions evolved against earlier generations.
3. Vulnerability profiles are category-dependent: copyright-category ASR is consistently highest across all generations under our 8B judge (though this specific number is sensitive to judge choice), Gemma 4 achieves 0% in three categories, but misinformation remains elevated (76.9%) even in the safest generation.

These results underscore the need for *adaptive, longitudinal* safety evaluation. Static benchmarks provide a snapshot; evolutionary probing with transfer analysis reveals the *trajectory* of safety alignment. We advocate for model developers to publish generational safety analyses alongside capability benchmarks, and for the research community to develop standardized protocols for cross-version safety comparison.

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# NeurIPS Paper Checklist

1. **Claims**
   (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]The abstract states two specific research questions (cross-generational transfer and non-monotonic robustness) and the introduction enumerates four concrete contributions, each tied to a results section.
2. **Limitations**
   (a) Does the paper discuss the limitations of the work performed by the authors? [Yes]Section 6 includes a dedicated Limitations paragraph covering: limited statistical power with 3 seeds, single model family, weak agreement between 8B and 70B judges on harm classification ($\kappa = 0.15$), HarmBench coverage bounds, evolutionary probe coverage, and text-only modality limitation.
3. **Theory Assumptions and Proofs**
   (a) For each theoretical result, does the paper provide the full set of assumptions and a complete (or correct) proof? [N/A]This is an empirical paper with no theoretical results.
4. **Experimental Result Reproducibility**
   (a) Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)? [Yes]Section 4 specifies all models, hyperparameters (grid dimensions, batch size, generations, concurrency), seed dataset (HarmBench 400 behaviors), judge model (Llama-3.1-8B-Instruct, self-hosted), serving infrastructure (vLLM 0.19.1, bf16, temperature 0), and hardware (H200, RTX 4090). The transfer replay protocol is fully described in Section 3.2.
5. **Open access to data and code**
   (a) Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material? [Yes]Anonymous code repository is provided as supplemental material, including the full evolution framework (Rust), analysis scripts (Python), orchestration scripts, and instructions for reproducing all experiments. Seed behaviors are from the public HarmBench dataset. Experiment result JSONs are included.
6. **Experimental Setting/Details**
   (a) Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results? [Yes]All MAP-Elites hyperparameters are specified in Section 4. No training is performed; we evaluate existing models. Evolution configuration: 200 generations, batch size 50, $6 \times 6 \times 8$ MAP-Elites grid, mutation rate 0.3, crossover rate 0.7.
7. **Experiment Statistical Significance**
   (a) Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments? [Yes]All experiments use 3 independent random seeds per model. We report mean $\pm$ standard deviation across 3 seeds. Paired bootstrap tests are used for pairwise model comparisons ($p < 0.05$ threshold). We acknowledge that $N = 3$ provides limited power for some contrasts.
8. **Experiments Compute Resources**
   (a) For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time for each experiment, etc.) needed to reproduce the experiments? [Yes]Section 4 reports: NVIDIA H200 (141 GB) for targets, RTX 4090 (24 GB) for the 8B judge, vLLM 0.19.1. Total GPU time: $\sim$60 hours (12 evolution runs at $\sim$4h each, 36 transfer replays at $\sim$8min each, 4 raw-seed baselines, 1 judge audit with Llama-3.1-70B on the H200).
9. **Code Of Ethics**

(a) Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics? [Yes]This work studies LLM safety alignment to improve defenses. We follow responsible disclosure practices: we do not release specific jailbreak prompts that could enable misuse, only aggregate statistics and methodology. The goal is to help model developers identify and fix vulnerabilities.

10. **Broader Impacts**

    (a) Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed? [Yes]Positive: enables systematic safety evaluation across model generations, helping developers prioritize alignment work. Negative: methodology could in principle be used to find and exploit vulnerabilities. We mitigate this by not publishing specific attack prompts and by framing the work as a defensive evaluation tool.

11. **Safeguards**

    (a) Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)? [Yes]We release aggregate statistics (transfer rates, ASR per category) and the evolution framework, but not the specific attack prompts or model responses from our experiments. The seed behaviors come from the publicly available HarmBench dataset.

12. **Licenses for existing assets**

    (a) Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected? [Yes]HarmBench is cited and used under its original license. Gemma models are used under Google's Gemma license. The judge model (Llama-3.1-8B-Instruct) is used under Meta's license.

13. **New Assets**

    (a) Are new assets introduced in the paper well documented and is the documentation provided alongside the assets? [Yes]The evolution framework is documented with README, CLI help text, and reproduction instructions. Analysis scripts include docstrings and usage examples.

14. **Crowdsourcing and Research with Human Subjects**

    (a) For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation? [N/A]No human subjects or crowdsourcing.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects**

    (a) Does the paper describe potential risks incurred by study participants, whether combative or non-combative to the subjects, and does it meet the necessary IRB criteria? [N/A]No human subjects.

16. **Declaration of LLM usage**

    (a) Does the paper describe the usage of LLMs in the research and/or writing process? [Yes]LLMs are used in two capacities: (1) as evaluation targets (the Gemma models being probed for safety), and (2) as a self-hosted automated judge (Llama-3.1-8B-Instruct scoring target responses). An AI coding assistant (Claude Code) was used to assist with analysis scripts, figure generation, and framework modifications.